

ML Studio - Versioning & Git Workflow

Current Features

- Email Verification
- Forgot Password / Reset Password
- Dataset Upload
- Dataset Preview
- Data Preprocessing
- Model Training
- Data Visualization (Pending)

Recommended Version Plan

v0.9.0-beta: Testing release with all current features except Data Visualization.

v1.0.0: First stable production release after completing and testing Data Visualization.

Future:

- v1.1.0: Model evaluation improvements, feature importance, model download.
- v1.2.0: Hyperparameter tuning, model comparison.
- v2.0.0: Major features/UI improvements and collaboration.

Git Branch Strategy

main -> Production

develop -> Testing/Staging

feature/* -> Individual feature branches

Development Workflow

1. Create feature branch from develop.
2. Develop locally.
3. Push feature branch.
4. Merge into develop.
5. Deploy Render Test services from develop.
6. Test thoroughly.
7. Merge develop into main.
8. Tag release (v1.0.0, v1.1.0, etc.).

Render Environments

Local: Local PostgreSQL + localhost.

Testing: Separate Render frontend, backend, and Neon test database.

Production: Separate Render frontend, backend, and Neon production database.

Upcoming Features

- Complete Data Visualization.
- Model Evaluation Dashboard.
- Feature Importance Charts.
- Download Trained Model.
- Hyperparameter Tuning.
- Model Comparison.
- UI/UX Improvements.

- Project History / Experiment Tracking.